INVARIANT AFFINE NEIGHBOURHOODS OF CODIMENSION-AT-MOST-ONE ORBITS IN A FINITE GROUPOID

THE STACKS PROJECT AUTHORS

Editorial note. No flatness assumption on s,t is added. The adjacent finite-flat affine-neighbourhood lemmas are only retained support.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Finite groupoid schemes, separated locally Noetherian schemes, affine neighbourhoods of finite codimension-one sets, finite-flat loci, integral modifications and invariant opens. The local finite-flat and affine-integral support is retained.

Source reference. Original lemma 0AB9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13012–13012. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05P8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7332–7332. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 054K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6643–6643. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AAW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12999–12999. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09NL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12219–12219. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01ZV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2610–2610. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:more-groupoids.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03JE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4636–4636. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01YQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2565–2565. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01OY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2211–2211. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 05YU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7668–7668. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AAU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12997–12997. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2487–2487. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00DS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 508–508. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03BH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4348–4348. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01K9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2042–2042. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0C3A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15185–15185. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01P9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2222–2222. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01KT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2062–2062. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01P8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2221–2221. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 09NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12221–12221. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01ZY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2613–2613. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0542: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6625–6625. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0903: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11396–11396. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let (U, R, s, t, c) be a groupoid scheme. Let $u \in U$. Assume*

(1) *s, t are finite morphisms,*
(2) *U is separated and locally Noetherian,*
(3) *$\dim(\mathcal{O}_{U,u'}) \leq 1$ for every point u' in the orbit of u.*

Then u is contained in an R-invariant affine open of U.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S. Assume s, t are finite. There exists an open subscheme $W \subset U$ and a closed subscheme $W' \subset W$ such that*

(1) *W and W' are R-invariant,*
(2) *$U = t(s^{-1}(\overline{W}))$ set theoretically,*
(3) *W is a thickening of W', and*

(4) *the maps s', t' of the restriction (W', R', s', t', c') are finite locally free.*

Proof. Consider the stratification $U = Z_0 \supset Z_1 \supset Z_2 \supset \ldots$ of Lemma 0AB9.

We will construct disjoint unions $W = \coprod_{r \geq 1} W_r$ and $W' = \coprod_{r \geq 1} W'_r$ with each $W'_r \to W_r$ a thickening of R-invariant subschemes of U such that the morphisms s'_r, t'_r of the restrictions $(W'_r, R'_r, s'_r, t'_r, c'_r)$ are finite locally free of rank r. To begin we set $W_1 = W'_1 = U \setminus Z_1$. This is an R-invariant open subscheme of U, it is true that W_0 is a thickening of W'_0, and the maps s'_1, t'_1 of the restriction $(W'_1, R'_1, s'_1, t'_1, c'_1)$ are isomorphisms, i.e., finite locally free of rank 1. Moreover, every point of $U \setminus Z_1$ is in $t(s^{-1}(\overline{W_1}))$.

Assume we have found subschemes $W'_r \subset W_r \subset U$ for $r \leq n$ such that

(1) $W_1, \ldots, W_n$ are disjoint,
(2) W_r and W'_r are R-invariant,
(3) $U \setminus Z_n \subset \bigcup_{r \leq n} t(s^{-1}(\overline{W_r}))$ set theoretically,
(4) W_r is a thickening of W'_r,
(5) the maps s'_r, t'_r of the restriction $(W'_r, R'_r, s'_r, t'_r, c'_r)$ are finite locally free of rank r.

Then we set

$$W_{n+1} = Z_n \setminus \Big(Z_{n+1} \cup \bigcup\nolimits_{r \leq n} t(s^{-1}(\overline{W_r})) \Big)$$

set theoretically and

$$W'_{n+1} = Z_n \setminus \Big(Z_{n+1} \cup \bigcup\nolimits_{r \leq n} t(s^{-1}(\overline{W_r})) \Big)$$

scheme theoretically. Then W_{n+1} is an R-invariant open subscheme of U because $Z_{n+1} \setminus \overline{U \setminus Z_{n+1}}$ is open in U and $\overline{U \setminus Z_{n+1}}$ is contained in the closed subset $\bigcup_{r \leq n} t(s^{-1}(\overline{W_r}))$ we are removing by property (3) and the fact that t is a closed morphism. It is clear that W'_{n+1} is a closed subscheme of W_{n+1} with the same underlying topological space. Finally, properties (1), (2) and (3) are clear and property (5) follows from Lemma 0AB9.

By Lemma 0AB9 we have $\bigcap Z_r = \emptyset$. Hence every point of U is contained in $U \setminus Z_n$ for some n. Thus we see that $U = \bigcup_{r \geq 1} t(s^{-1}(\overline{W_r}))$ set theoretically and we see that (2) holds. Thus $W' \subset W$ satisfy (1), (2), (3), and (4). □

Let (U, R, s, t, c) be a groupoid scheme. Given a point $u \in U$ the *R-orbit* of u is the subset $t(s^{-1}(\{u\}))$ of U.

Lemma 0.3. *In Lemma 0.2 assume in addition that s and t are of finite presentation. Then*

(1) *the morphism $W' \to W$ is of finite presentation, and*
(2) *if $u \in U$ is a point whose R-orbit consists of generic points of irreducible components of U, then $u \in W$.*

Proof. In this case the stratification $U = Z_0 \supset Z_1 \supset Z_2 \supset \ldots$ of Lemma 0AB9 is given by closed immersions $Z_k \to U$ of finite presentation, see Divisors, Lemma 05P8. Part (1) follows immediately from this as $W' \to W$ is locally given by intersecting the open W by Z_r. To see part (2) let $\{u_1, \ldots, u_n\}$ be the orbit of u. Since the closed subschemes Z_k are R-invariant and $\bigcap Z_k = \emptyset$, we find an k such that $u_i \in Z_k$ and $u_i \notin Z_{k+1}$ for all i. The image of $Z_k \to U$ and $Z_{k+1} \to U$ is locally constructible (Morphisms, Theorem 054K). Since $u_i \in U$ is a generic point of an irreducible component of U, there exists an open neighbourhood U_i of u_i which is contained in $Z_k \setminus Z_{k+1}$ set theoretically (Properties, Lemma 0AAW). In the proof of Lemma 0.2 we have constructed W as a disjoint union $\coprod W_r$ with $W_r \subset Z_{r-1} \setminus Z_r$ such that $U = \bigcup t(s^{-1}(\overline{W_r}))$. As $\{u_1, \ldots, u_n\}$ is an R-orbit we see

that $u \in t(s^{-1}(\overline{W_r}))$ implies $u_i \in \overline{W_r}$ for some i which implies $U_i \cap W_r \neq \emptyset$ which implies $r = k$. Thus we conclude that u is in

$$W_{k+1} = Z_k \setminus \left(Z_{k+1} \cup \bigcup\nolimits_{r \leq k} t(s^{-1}(\overline{W_r})) \right)$$

as desired. □

Lemma 0.4. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S. Assume s, t are finite and of finite presentation and U quasi-separated. Let $u_1, \ldots, u_m \in U$ be points whose orbits consist of generic points of irreducible components of U. Then there exist R-invariant subschemes $V' \subset V \subset U$ such that*

(1) *$u_1, \ldots, u_m \in V'$,*
(2) *V is open in U,*
(3) *V' and V are affine,*
(4) *$V' \subset V$ is a thickening of finite presentation,*
(5) *the morphisms s', t' of the restriction (V', R', s', t', c') are finite locally free.*

Proof. Let $W' \subset W \subset U$ be as in Lemma 0.2. By Lemma 0.3 we get $u_j \in W$ and that $W' \to W$ is a thickening of finite presentation. By Limits, Lemma 09NL it suffices to find an R-invariant affine open subscheme V' of W' containing u_j (because then we can let $V \subset W$ be the corresponding open subscheme which will be affine). Thus we may replace (U, R, s, t, c) by the restriction (W', R', s', t', c') to W'. In other words, we may assume we have a groupoid scheme (U, R, s, t, c) whose morphisms s and t are finite locally free. By Properties, Lemma 01ZV we can find an affine open containing the union of the orbits of $u_1, \ldots, u_m$. Finally, we can apply Groupoids, Lemma 03JE to conclude. □

The following lemma is a special case of Lemma 0.4 but we redo the argument as it is slightly easier in this case (it avoids using Lemma 0.3).

Lemma 0.5. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S. Assume s, t finite, U is locally Noetherian, and $u_1, \ldots, u_m \in U$ points whose orbits consist of generic points of irreducible components of U. Then there exist R-invariant subschemes $V' \subset V \subset U$ such that*

(1) *$u_1, \ldots, u_m \in V'$,*
(2) *V is open in U,*
(3) *V' and V are affine,*
(4) *$V' \subset V$ is a thickening,*
(5) *the morphisms s', t' of the restriction (V', R', s', t', c') are finite locally free.*

Proof. Let $\{u_{j1}, \ldots, u_{jn_j}\}$ be the orbit of u_j. Let $W' \subset W \subset U$ be as in Lemma 0.2. Since $U = t(s^{-1}(\overline{W}))$ we see that at least one $u_{ji} \in \overline{W}$. Since u_{ji} is a generic point of an irreducible component and U locally Noetherian, this implies that $u_{ji} \in W$. Since W is R-invariant, we conclude that $u_j \in W$ and in fact the whole orbit is contained in W. By Cohomology of Schemes, Lemma 01YQ it suffices to find an R-invariant affine open subscheme V' of W' containing $u_1, \ldots, u_m$ (because then we can let $V \subset W$ be the corresponding open subscheme which will be affine). Thus we may replace (U, R, s, t, c) by the restriction (W', R', s', t', c') to W'. In other words, we may assume we have a groupoid scheme (U, R, s, t, c) whose morphisms s and t are finite locally free. By Properties, Lemma 01ZV we can find an affine open containing $\{u_{ij}\}$ (a locally Noetherian scheme is quasi-separated by Properties, Lemma 01OY). Finally, we can apply Groupoids, Lemma 03JE to conclude. □

Lemma 0.6. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S with s, t integral. Let $g : U' \to U$ be an integral morphism such that every R-orbit in U meets $g(U')$. Let (U', R', s', t', c') be the restriction of R to U'. If $u' \in U'$ is*

contained in an R'-invariant affine open, then the image $u \in U$ is contained in an R-invariant affine open of U.

Proof. Let $W' \subset U'$ be an R'-invariant affine open. Set $\tilde{R} = U' \times_{g,U,t} R$ with maps $\mathrm{pr}_0 : \tilde{R} \to U'$ and $h = s \circ \mathrm{pr}_1 : \tilde{R} \to U$. Observe that pr_0 and h are integral. It follows that $\tilde{W} = \mathrm{pr}_0^{-1}(W')$ is affine. Since W' is R'-invariant, the image $W = h(\tilde{W})$ is set theoretically R-invariant and $\tilde{W} = h^{-1}(W)$ set theoretically (details omitted). Thus, if we can show that W is open, then W is a scheme and the morphism $\tilde{W} \to W$ is integral surjective which implies that W is affine by Limits, Proposition 05YU. However, our assumption on orbits meeting U' implies that $h : \tilde{R} \to U$ is surjective. Since an integral surjective morphism is submersive (Topology, Lemma 0AAU and Morphisms, Lemma 01WM) it follows that W is open. □

The following technical lemma produces "almost" invariant functions in the situation of a finite groupoid on a quasi-affine scheme.

Lemma 0.7. *Let (U, R, s, t, c) be a groupoid scheme with s, t finite and of finite presentation. Let $u_1, \ldots, u_m \in U$ be points whose R-orbits consist of generic points of irreducible components of U. Let $j : U \to \mathrm{Spec}(A)$ be an immersion. Let $I \subset A$ be an ideal such that $j(U) \cap V(I) = \emptyset$ and $V(I) \cup j(U)$ is closed in $\mathrm{Spec}(A)$. Then there exists an $h \in I$ such that $j^{-1}D(h)$ is an R-invariant affine open subscheme of U containing $u_1, \ldots, u_m$.*

Proof. Let $u_1, \ldots, u_m \in V' \subset V \subset U$ be as in Lemma 0.4. Since $U \setminus V$ is closed in U, j an immersion, and $V(I) \cup j(U)$ is closed in $\mathrm{Spec}(A)$, we can find an ideal $J \subset I$ such that $V(J) = V(I) \cup j(U \setminus V)$. For example we can take the ideal of elements of I which vanish on $j(U \setminus V)$. Thus we can replace (U, R, s, t, c), $j : U \to \mathrm{Spec}(A)$, and I by (V', R', s', t', c'), $j|_{V'} : V' \to \mathrm{Spec}(A)$, and J. In other words, we may assume that U is affine and that s and t are finite locally free. Take any $f \in I$ which does not vanish at all the points in the R-orbits of $u_1, \ldots, u_m$ (Algebra, Lemma 00DS). Consider

$$g = \mathrm{Norm}_s(t^\sharp(j^\sharp(f))) \in \Gamma(U, \mathcal{O}_U)$$

Since $f \in I$ and since $V(I) \cup j(U)$ is closed we see that $U \cap D(f) \to D(f)$ is a closed immersion. Hence $f^n g$ is the image of an element $h \in I$ for some $n > 0$. We claim that h works. Namely, we have seen in Groupoids, Lemma 03BH that g is an R-invariant function, hence $D(g) \subset U$ is R-invariant. Since f does not vanish on the orbit of u_j, the function g does not vanish at u_j. Moreover, we have $V(g) \supset V(j^\sharp(f))$ and hence $j^{-1}D(h) = D(g)$. □

Lemma 0.8. *Let (U, R, s, t, c) be a groupoid scheme. If s, t are finite, and $u, u' \in R$ are distinct points in the same orbit, then u' is not a specialization of u.*

Proof. Let $r \in R$ with $s(r) = u$ and $t(r) = u'$. If $u \rightsquigarrow u'$ then we can find a nontrivial specialization $r \rightsquigarrow r'$ with $s(r') = u'$, see Schemes, Lemma 01K9. Set $u'' = t(r')$. Note that $u'' \neq u'$ as there are no specializations in the fibres of a finite morphism. Hence we can continue and find a nontrivial specialization $r' \rightsquigarrow r''$ with $s(r'') = u''$, etc. This shows that the orbit of u contains an infinite sequence $u \rightsquigarrow u' \rightsquigarrow u'' \rightsquigarrow \ldots$ of specializations which is nonsense as the orbit $t(s^{-1}(\{u\}))$ is finite. □

Lemma 0.9. *Let $j : V \to \mathrm{Spec}(A)$ be a quasi-compact immersion of schemes. Let $f \in A$ be such that $j^{-1}D(f)$ is affine and $j(V) \cap V(f)$ is closed. Then V is affine.*

Proof. This follows from Morphisms, Lemma 0C3A but we will also give a direct proof. Let $A' = \Gamma(V, \mathcal{O}_V)$. Then $j' : V \to \mathrm{Spec}(A')$ is a quasi-compact open immersion, see Properties, Lemma 01P9. Let $f' \in A'$ be the image of f. Then

$(j')^{-1}D(f') = j^{-1}D(f)$ is affine. On the other hand, $j'(V) \cap V(f')$ is a subscheme of $\mathrm{Spec}(A')$ which maps isomorphically to the closed subscheme $j(V) \cap V(f)$ of $\mathrm{Spec}(A)$. Hence it is closed in $\mathrm{Spec}(A')$ for example by Schemes, Lemma 01KT. Thus we may replace A by A' and assume that j is an open immersion and $A = \Gamma(V, \mathcal{O}_V)$.

In this case we claim that $j(V) = \mathrm{Spec}(A)$ which finishes the proof. If not, then we can find a principal affine open $D(g) \subset \mathrm{Spec}(A)$ which meets the complement and avoids the closed subset $j(V) \cap V(f)$. Note that j maps $j^{-1}D(f)$ isomorphically onto $D(f)$, see Properties, Lemma 01P8. Hence $D(g)$ meets $V(f)$. On the other hand, $j^{-1}D(g)$ is a principal open of the affine open $j^{-1}D(f)$ hence affine. Hence by Properties, Lemma 01P8 again we see that $D(g)$ is isomorphic to $j^{-1}D(g) \subset j^{-1}D(f)$ which implies that $D(g) \subset D(f)$. This contradiction finishes the proof. □

Proof. The R-orbit of u is finite. By conditions (2) and (3) it is contained in an affine open U' of U, see Varieties, Proposition 09NN. Then $t(s^{-1}(U \setminus U'))$ is an R-invariant closed subset of U which does not contain u. Thus $U \setminus t(s^{-1}(U \setminus U'))$ is an R-invariant open of U' containing u. Replacing U by this open we may assume U is quasi-affine.

By Lemma 0.6 we may replace U by its reduction and assume U is reduced. This means R-invariant subschemes $W' \subset W \subset U$ of Lemma 0.2 are equal $W' = W$. As $U = t(s^{-1}(\overline{W}))$ some point u' of the R-orbit of u is contained in $\overline{W}$ and by Lemma 0.6 we may replace U by $\overline{W}$ and u by u'. Hence we may assume there is a dense open R-invariant subscheme $W \subset U$ such that the morphisms s_W, t_W of the restriction (W, R_W, s_W, t_W, c_W) are finite locally free.

If $u \in W$ then we are done by Groupoids, Lemma 03JE (because W is quasi-affine so any finite set of points of W is contained in an affine open, see Properties, Lemma 01ZY). Thus we assume $u \notin W$ and hence none of the points of the orbit of u is in W. Let $\xi \in U$ be a point with a nontrivial specialization to a point u' in the orbit of u. Since there are no specializations among the points in the orbit of u (Lemma 0.8) we see that ξ is not in the orbit. By assumption (3) we see that ξ is a generic point of U and hence $\xi \in W$. As U is Noetherian there are finitely many of these points $\xi_1, \ldots, \xi_m \in W$. Because s_W, t_W are flat the orbit of each ξ_j consists of generic points of irreducible components of W (and hence U).

Let $j : U \to \mathrm{Spec}(A)$ be an immersion of U into an affine scheme (this is possible as U is quasi-affine). Let $J \subset A$ be an ideal such that $V(J) \cap j(W) = \emptyset$ and $V(J) \cup j(W)$ is closed. Apply Lemma 0.7 to the groupoid scheme (W, R_W, s_W, t_W, c_W), the morphism $j|_W : W \to \mathrm{Spec}(A)$, the points ξ_j, and the ideal J to find an $f \in J$ such that $(j|_W)^{-1}D(f)$ is an R_W-invariant affine open containing ξ_j for all j. Since $f \in J$ we see that $j^{-1}D(f) \subset W$, i.e., $j^{-1}D(f)$ is an R-invariant affine open of U contained in W containing all ξ_j.

Let Z be the reduced induced closed subscheme structure on

$$U \setminus j^{-1}D(f) = j^{-1}V(f).$$

Then Z is set theoretically R-invariant (but it may not be scheme theoretically R-invariant). Let (Z, R_Z, s_Z, t_Z, c_Z) be the restriction of R to Z. Since $Z \to U$ is finite, it follows that s_Z and t_Z are finite. Since $u \in Z$ the orbit of u is in Z and agrees with the R_Z-orbit of u viewed as a point of Z. Since $\dim(\mathcal{O}_{U,u'}) \leq 1$ and since $\xi_j \notin Z$ for all j, we see that $\dim(\mathcal{O}_{Z,u'}) \leq 0$ for all u' in the orbit of u. In other words, the R_Z-orbit of u consists of generic points of irreducible components of Z.

Let $I \subset A$ be an ideal such that $V(I) \cap j(U) = \emptyset$ and $V(I) \cup j(U)$ is closed. Apply Lemma 0.7 to the groupoid scheme (Z, R_Z, s_Z, t_Z, c_Z), the restriction $j|_Z$, the ideal I, and the point $u \in Z$ to obtain $h \in I$ such that $j^{-1}D(h) \cap Z$ is an R_Z-invariant open affine containing u.

Consider the R_W-invariant (Groupoids, Lemma 03BH) function

$$g = \mathrm{Norm}_{s_W}(t_W^\sharp(j^\sharp(h)|_W)) \in \Gamma(W, \mathcal{O}_W)$$

(In the following we only need the restriction of g to $j^{-1}D(f)$ and in this case the norm is along a finite locally free morphism of affines.) We claim that

$$V = (W_g \cap j^{-1}D(f)) \cup (j^{-1}D(h) \cap Z)$$

is an R-invariant affine open of U which finishes the proof of the lemma. It is set theoretically R-invariant by construction. As V is a constructible set, to see that it is open it suffices to show it is closed under generalization in U (Topology, Lemma 0542 or the more general Topology, Lemma 0903). Since $W_g \cap j^{-1}D(f)$ is open in U, it suffices to consider a specialization $u_1 \rightsquigarrow u_2$ of U with $u_2 \in j^{-1}D(h) \cap Z$. This means that h is nonzero in $j(u_2)$ and $u_2 \in Z$. If $u_1 \in Z$, then $j(u_1) \rightsquigarrow j(u_2)$ and since h is nonzero in $j(u_2)$ it is nonzero in $j(u_1)$ which implies $u_1 \in V$. If $u_1 \notin Z$ and also not in $W_g \cap j^{-1}D(f)$, then $u_1 \in W$, $u_1 \notin W_g$ because the complement of $Z = j^{-1}V(f)$ is contained in $W \cap j^{-1}D(f)$. Hence there exists a point $r_1 \in R$ with $s(r_1) = u_1$ such that h is zero in $t(r_1)$. Since s is finite we can find a specialization $r_1 \rightsquigarrow r_2$ with $s(r_2) = u_2$. However, then we conclude that h is zero in $u_2' = t(r_2)$ which contradicts the fact that $j^{-1}D(h) \cap Z$ is R-invariant and u_2 is in it. Thus V is open.

Observe that $V \subset j^{-1}D(h)$ for our function $h \in I$. Thus we obtain an immersion

$$j' : V \longrightarrow \mathrm{Spec}(A_h)$$

Let $f' \in A_h$ be the image of f. Then $(j')^{-1}D(f')$ is the principal open determined by g in the affine open $j^{-1}D(f)$ of U. Hence $(j')^{-1}D(f)$ is affine. Finally, $j'(V) \cap V(f') = j'(j^{-1}D(h) \cap Z)$ is closed in $\mathrm{Spec}(A_h/(f')) = \mathrm{Spec}((A/f)_h) = D(h) \cap V(f)$ by our choice of $h \in I$ and the ideal I. Hence we can apply Lemma 0.9 to conclude that V is affine as claimed above. $\square$